

Shoes or no shoes during locomotion and exercise: Training potential for fascial structures of the lower extremity

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Introduction

It is one of the most controversial and vivid discussions in the world of sport and exercise: Should I wear “regular” sports shoes, shoes with added material components, shoes with reduced material components, or no shoes at all during locomotion? While wearing shoes is generally considered to add a certain functionality, it may also prevent the foot from unfolding its full natural potential based on its sophisticated anatomy. Furthermore, wearing shoes may even prevent the foot from receiving beneficial training stimuli suited to enhance its functional capability. This chapter focuses on the effect of wearing shoes, or not, during locomotion and exercise. Until now, these considerations have been widely linked to movement kinetics and kinematics, as well as aspects of muscle activity. However, they have not been linked to the structural disposition or the functional performance of the fascial network or the individual fasciae of the lower extremity. As the role of fascia for human performance and well-being draws increasing attention, it is beneficial to understand how the fascial tissue of the lower extremity is affected by different types of shoes (or not) during activities of daily life, athletic training or physical therapy. Recommendations follow how to implement specific stimulation of fascial tissue of the lower extremity during locomotion and exercise training regimens of individuals, athletes and patients. These should aid interested individuals,

as well as coaches and therapists, in considering another worthwhile component for their training or rehabilitation regimens.

Why athletic shoes are worn

Shoes form an artificial interface between the foot and the ground. Thereby, they influence the way people walk, run, or execute other types of locomotion and exercise. Importantly, not only the shoe, but also the ground can be varied during locomotion and exercise by using different surfaces present in natural environments, or by arranging different surface conditions in specific athletic conditioning or therapeutic environments. The whole body, but in particular the foot and the lower extremity, is affected by the combination of shoe and surface, which determine the functional circumstances individuals, athletes and patients cope with when contacting the ground during locomotion and exercise. While shoes and surfaces may be easily changed, the foot and the lower extremity with their formed anatomical structures and functional characteristics need to be trained over a longer period of time. Bio-positive stimulation eventually leads to a higher structural and functional level, while bio-negative stimulation, due to acute or accumulated overloading, eventually leads to trauma or overuse injuries.

For a number of sports or disciplines, shoes are an inherent necessity or provide an obvious advantage for their execution and related

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performance. Examples include ice hockey, soccer, or track and field sprinting. In activities where shoes are not a necessity, however, what are the reasons an artificial shoe interface is preferred over the more direct foot-to-ground contact experienced being barefoot?

The four core functional aspects, comfort, performance, injury prevention and training, sum up the main reasons for wearing athletic shoes during locomotion and exercise (Figure 32.1). Whereas comfort and performance provide instantaneous benefits to the individual, injury prevention has an acute component, when supporting the avoidance of traumatic injuries, and a remote component, when supporting the avoidance of overuse injuries. Additionally, certain shoes are regarded as specific training devices, when designed to provide enhanced stimuli for a bio-positive development of the musculoskeletal system of the wearer, which is not provided by regular athletic shoes. In this sense, barefoot, as the no-shoe condition, may be regarded as a training option, too.

When referring to athletic shoes as training devices, two well researched and also extensively marketed concepts are footwear minimization (Sinclair, 2014) and footwear instability (Apps et al., 2017) (Figures 32.2 and 32.3). Footwear instability concepts commonly feature additions to or variation of the shoe midsole and outsole construction, to create functional instability during the stance phase of locomotion and exercise. Thereby, it is intuitive that actual performance when wearing less stable shoes is decreased as a trade-off in creating increased training stimuli compared with regular athletic shoes. Footwear minimization concepts commonly feature a reduction of shoe material, involving components of the shoe upper and the shoe sole. This results in lower shoe weight but also in reduced comfort, mainly due to decreased cushioning, while on the other hand it induces more lateral stability, by bringing the foot closer to the ground, thereby reducing the lateral lever arm. Furthermore, footwear minimization concepts free the foot to a certain degree, allowing the numerous foot segments to act more individually, fostering pronounced

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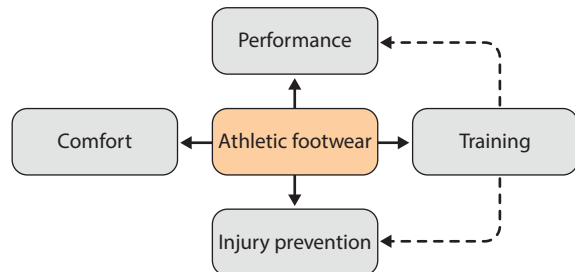


Figure 32.1

Core functions of athletic footwear, enhanced from Sterzing (2015), emphasizing the influence of the training function on functions of performance and injury prevention.

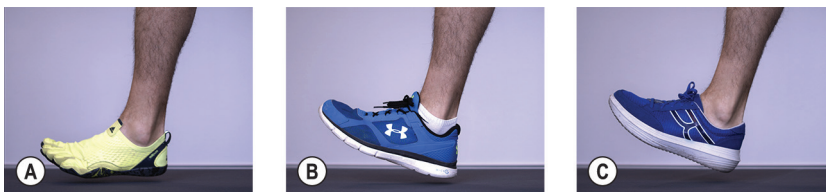


Figure 32.2

Examples of differing shoe concepts from various brands during running, ~20–30 ms before ground contact: (A) minimalistic footwear, (B) regular running footwear and (C) elastic instability footwear.

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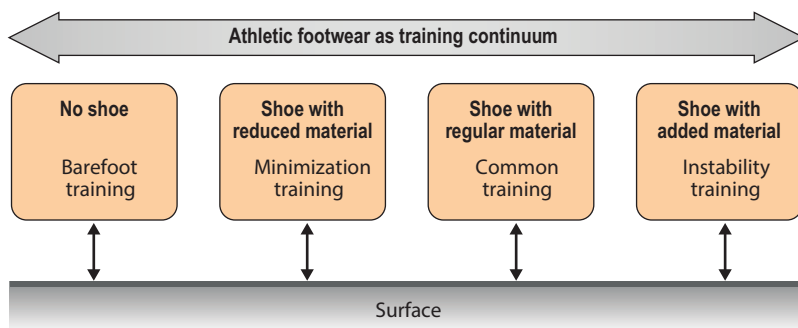


Figure 32.3

Material aspect of athletic shoes from a training shoe continuum perspective.

gripping of the ground. In this sense it is argued that the more “minimal” our footwear, the more we expose the foot to its sophisticated natural potential and therefore improve its functionality.

The current debate about the benefits of wearing athletic shoes, or not, covers direct biomechanical effects during locomotion and exercise. It also addresses if and how such biomechanical adaptations affect anatomical and physiological structures of the foot and the lower extremity in the mid and long term, in the sense of functional training. With regard to training effects due to the wearing of shoes, to date the structural and functional characteristics of fascia tissue have been ignored.

Biomechanical effects of barefoot, minimization and instability movement

In order to delve into the potential consequences of biomechanical adaptations to the shoe and surface type on fascial tissue of the lower extremity, we need to illustrate the main biomechanical alterations from barefoot, minimization, and instability locomotion and exercise. The effects are explained in comparison to locomotion and exercise using regular athletic shoes, as they resemble the “habitual” standard. While most of the available insights are related to analyses of walking and running, gym class or

dance movements should also be considered, as they often include variations of walking and running, like lunges, or steps and jumps in various directions.

Barefoot and minimization locomotion and exercise

During shod walking and shod endurance running, the rear-foot strike is the overwhelmingly applied foot-strike pattern. In walking, the rear-foot strike is widely maintained irrespective of shoe or surface type, as forces and regional pressures are comparatively low, and thus discomfort and pain levels are bearable in most scenarios (Fong Yan et al., 2013), even though they are reported to be close to the individuals’ pain thresholds (Wearing et al., 2014). In running, the rear-foot strike is often changed to a more moderate rear-foot strike and, quite frequently, even to a mid-foot or fore-foot strike when running barefoot or wearing minimization shoes, as forces and regional pressures are twice as high in heel striking than in fore-foot striking (Shih et al., 2013). Adopting a mid-foot or fore-foot strike pattern during barefoot or minimization running is considered a balance between avoiding high pressure under the heel and bearing higher energetic expenditure. Consequently, a consistent application of fore-foot strike during barefoot or minimization running is rarely

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seen among general individuals. Differences in ground pressures for running with and without shoes are displayed, allowing visualization of the altered ground contact development between running barefoot and in regular shoes at moderate speed (Figure 32.4).

To avoid increased discomfort and pain during barefoot or minimization running, runners adapt their foot-strike patterns in an individual manner. Some apply a more moderate rear-foot strike, some use a mid-foot strike, while some exhibit a fore-foot strike. While these terms seem discrete in nature, we need to be aware that within the range of a pronounced rear-foot strike and a pronounced fore-foot strike, numerous variations of strike-type execution may take place, depending on the runner. Interestingly, changes in foot-strike behavior can also be executed purely voluntarily for the purpose of variation of training regimens, which becomes important when aiming to induce certain training stimuli for the fascial network of the lower

extremity. Due to the strike type applied, the force transmission from the ground to the foot, and on to the lower extremity, changes its nature influencing respective effects on the muscular-skeletal system, as well as on the fascia network of the lower extremity. Whenever impacts are created by ground contacts, or propulsion forces need to be transmitted through the foot, these principles may be applied as they result in similar consequences for the tissues of the leg, and therefore on the fascia network (Figure 32.5). Being aware of these principles allows strike-type variation to be systematically implemented into training regimens by the individual, the coach, and the therapist.

Instability locomotion and exercise

Instability locomotion and exercise can be executed by wearing instability shoes during both everyday and athletic training movements, or by just executing such movements on unstable surfaces. Therefore, the instability induced can

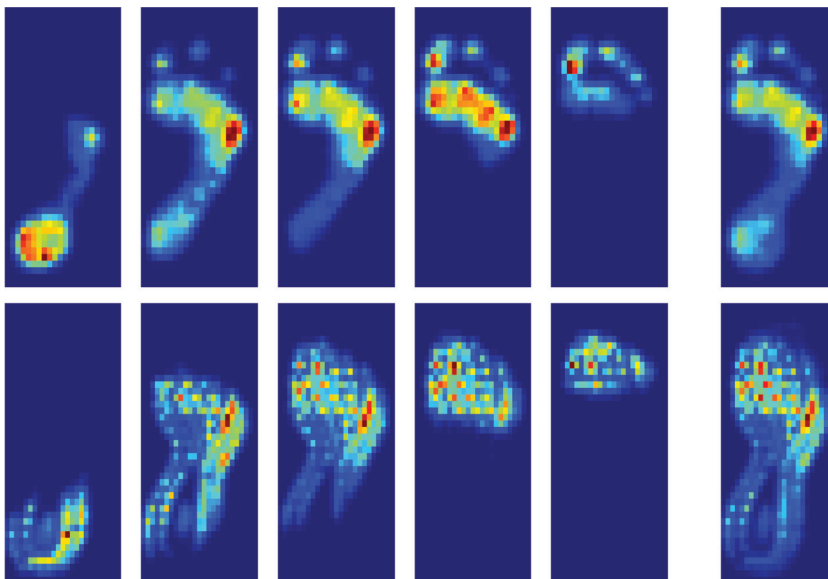


Figure 32.4

Ground pressures during barefoot (top) and shod (bottom) endurance running at moderate speed on a hard surface at 10%, 30%, 50%, 70% and 90% of stance (left), and peak pressures across stance (right). Barefoot running shows a less pronounced rather moderate rear-foot strike and an earlier foot flat during stance, while exhibiting high localized peak pressures across the foot sole, due to the anatomical foot structure not being shielded by the shoe. Shod running ground pressures are smoothed due to being distributed across the whole shoe sole.

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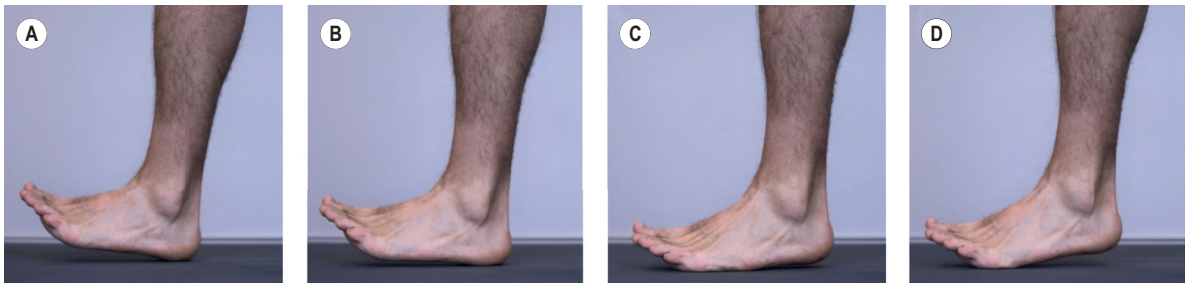


Figure 32.5

Initial ground contact during barefoot running with differing foot-strike patterns to illustrate the dominant ground reaction force attack location during (A) pronounced rear-foot strike, (B) moderate rear-foot strike, (C) mid-foot strike and (D) fore-foot strike.

have different forms. The basic form is referred to as constant instability. This is induced by shoes that do not change their instability properties, such as those featuring an outsole rocker or a soft outsole padding, and on surfaces that have consistent properties, like soft mats or sand. The advanced form can be referred to as varied or random instability. This may also be induced by shoes, artificial treadmill surfaces or natural surfaces, like trails. A commonly accepted effect of instability shoes is an enhanced activation of the small muscles surrounding the ankle, in response to ankle instability induced during exercise (Landry et al., 2010). Frequent use of instability shoes has also indicated an increased amount of movement variability: a factor considered beneficial, as it requires the wearer to negotiate different types of movement execution. However, it has also been shown that such movement variability decreases as wearers become increasingly familiarized with this constant instability stimulus (Stöggel et al., 2010). As a result, varied instability movements are endorsed, by proposing random instability footwear, or by movement execution on artificial or natural instability surfaces, for example, specifically adapted treadmills or trails. This means the individual does not become familiar

with the instability stimuli induced (Apps et al., 2017).

Mechanisms of fascia stimulation and fascia response

Probable underlying mechanisms to be considered when addressing the effect of barefoot locomotion and exercise (as well as wearing minimization or instability shoes during locomotion and exercise, and when executing locomotion and exercise on instability surfaces) might be:

- Distal shift of plantar force during foot strike, causing increased fascia stimulation
- Increased foot segment activity, causing increased muscle, tendon, and fascia stimulation
- Increased ankle instability, causing increased muscle, tendon, and fascia stimulation.

Distal shift of plantar force

As illustrated in Figure 32.5, the plantar force attack location during pronounced rear-foot strikes and during moderate rear-foot strikes is beneath the heel. This results in predominantly bony force transmission, primarily originating at

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the calcaneus, leaving the fascial network of the lower extremity to some extent at rest. This scenario is changed when the plantar force vector attack location shifts distally, under the mid-foot during mid-foot strikes, or under the balls of the feet during fore-foot strikes. In these cases, which can be fostered by barefoot locomotion or minimization shoe running, increased loading and tensile stretch of muscles, tendons and fasciae on the plantar side of the foot occurs, as well as of the posterior side of the shank and of the knee extensors. In these areas, the higher stress during foot strike needs to be absorbed by the muscles, tendons and fascia, and can therefore be considered as an eccentric training of these structures. Under such circumstances, the myofascial structures lengthen during eccentric muscle contraction due to high external loads. By contrast, shod running predominantly fosters rear-foot or moderate rear-foot striking and, thus, the loading is primarily transferred through bones and absorbed by cartilage within the ankle and the knee joint. It is concluded that foot-strike alteration modifies the stimulation of the fascial network when the plantar force vector attack location is shifted from the rear-foot to the mid-foot or onto the fore-foot. Such a shift can be expected to occur more often when reducing shoe material or during barefoot locomotion and exercise.

Increased foot segment activity

Barefoot and minimization footwear locomotion increases the mobility of individual foot segments which, in turn, increases the activity of intersegmental muscles and connective tissues of the foot. While this intersegmental mobility causes extra activity of these structures during the complete stance phase, it is of specific importance during terminal stance until push-off. During this phase, forward and upward propulsion is generated. To transfer this force, which is generated at the ankle joint and transmitted

towards the fore-foot region, the foot needs to stiffen. During barefoot locomotion, this stiffness is mainly produced by the intrinsic foot muscles and the plantar fascia, whereas during shod locomotion the stiffness of the foot/shoe unit is partially provided by the shoe. It is assumed that the stiffer the shoe construction the less stiffness needs to be generated by the foot structures, thus reducing the physiological incentive for the muscle and fascia tissue to be heavily involved. Consequently, it is concluded that long-term shod walking and shod running does not train intrinsic foot muscles and connective tissues as much as barefoot locomotion does, thus also having less beneficial effects on the functional structure of the foot arch (Hollander et al., 2017)

Increased ankle instability

Changes in ankle stability and related muscle, tendon, and fascia stimulation are well investigated for the muscle tendon complex during standing, walking and running (Apps et al., 2017). It is concluded that a decrease of ankle stability induced by wearing instability footwear benefits the stimulation of the fascial network around the ankle joint and the shank, alongside the previously mentioned stimulation of the small muscles of the lower extremity around the ankle joint.

Footwear training effects on the general fascial network and specific fasciae

As with all training, structural and functional adaptations of the bony and the musculoskeletal system, due to locomotion and exercise, need time to develop, as will potential structural and functional fascial adaptations. General effects on the fascia network are described below, followed by effects on specific fascial structures. At this point, such adaptations are only considered

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potentially, as sound scientific evidence of the proposed adaptations is not available yet.

Footwear effects on the general fascia network

In the following, wearing of shoes or being bare-foot is analyzed for the effects of fascial structure and function. Biomechanical adaptations in response to differing loading scenarios show some already well-investigated effects on bones, ligaments, muscles and tendons. Two comprehensive reviews (Guadalupe-Grau et al., 2009; Karlsson et al., 2008) demonstrate that high-impact physical activities have higher osteogenic (bone-building) effects than low-impact physical activities, such as swimming. While less investigated, it seems reasonable that fascial tissue is also affected by these adaptations as the above-mentioned structures are permeated by the fascial network: the number of collagen fibers in fasciae, and also their orientation, is highly adaptable and reacts directly to applied loads. Consequently, in fasciae that are regularly exposed to increased loading, the number of collagen fibers along the load axis is increased and their orientation is in better alignment with the predominantly applied force vectors, whereas fibers of less mobilized fasciae show a markedly irregular orientation with respect to the predicted most important force vectors in locomotion (Stecco, 2015). Despite these insights into fascial structure and functionality, the complexity of the fascial network and the wide variation of fascial tissue have impeded sound scientific evidence regarding the relationship of physiological fascial structure in response to the loads applied.

Footwear effects on specific fasciae

Alongside the effects on the general fascia network, focusing on specific fascial structures allows further potential effects to be proposed.

In particular, specific fascial impairments are addressed as locomotion and exercise form a substantial part of prevention and rehabilitation regimens. While such regimens routinely consider the individual circumstances of a patient, reflected in the therapeutic motor tasks and their doses, little attention is awarded to the aspect of the shoe type worn during the execution of such prevention and rehabilitation sessions.

Plantar fascia

Two different fascia types, superficial and deep, are referred to. The network of the deep fasciae of the foot and shank are of particular importance functionally. The plantar fascia (aponeurosis plantaris) resembles a fusion of superficial and deep fascial layers. The combination of these two layers results in a fascial structure exhibiting high stiffness, due to the functional alignment of the deep-layer fibers in longitudinal foot direction, alongside high proprioceptive capacity, due to its connection to the skin of its superficial layer fibers. The plantar fascia is a key structure as it spans over the foot's whole plantar side, connecting the tuberosity of the heel bone (calcaneus) and the heads of the metatarsal bones, the balls of the foot. In this sense, the plantar fascia is strongly exposed to the ground reaction forces experienced during foot strike and push-off.

Several functions are attributed to the plantar fascia. Its main function is the stabilization of the longitudinal foot arch by absorbing and distributing loads along the longitudinal axis of the foot. Hicks (1954) has described it as beam and truss, as it is subject to bending during propulsion while absorbing energy during landing and over general stance (Benjamin, 2009). The fore-foot strike pattern uses this load absorption and propulsion mechanism of the plantar fascia for transferring loads through the longitudinal axis of the foot. Several investigations link exposure of the plantar fascia to high loading with an

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increase of its stiffness, thus strengthening the longitudinal arch of the foot. It therefore is assumed that by fostering mid- or fore-foot strike patterns, running barefoot or with minimization footwear stiffens the plantar fascia, and consequently has stabilizing effects of the longitudinal arch of the foot. A study comparing the foot arch of 425 habitually shod children with 385 habitually barefoot children supports this assumption by observing higher foot arches in habitually barefoot children (Hollander et al., 2017).

High loading and lengthening stimuli of the plantar fascia are reported to have bio-positive effects in the treatment of plantar fasciitis, a common inflammatory pathology among athletes, usually experienced at the transition area of the plantar fascia and the calcaneus. This is interesting in the context of barefoot versus shod running, as the fore-foot strike pattern induced by running barefoot, or applied voluntarily during shod running, corresponds to eccentric training of the intrinsic foot muscles and the plantar fascia. Whether barefoot running should be applied as a therapy in people suffering from plantar fasciitis is, however, controversially discussed and extreme views on both side of the spectrum are put forward. Traditionally, people suffering from plantar fasciitis are told to avoid walking and running barefoot on hard surfaces as the high loads might increase the inflammatory response, whereas barefoot promoters assume that arch strengthening will eventually reduce stress on the plantar fascia and cite anecdotal evidence of reduced injuries in barefoot runners (Lieberman et al., 2010). Thus, further prospective investigations are needed to provide solid recommendations in this regard.

Achilles tendon

A further aspect of the plantar fascia needs to be mentioned in the context of potential footwear

influence. There is compelling evidence that the plantar fascia is continuous with the Achilles tendon and, thus, not only transfers stress and loading between hind and fore-foot, but is also involved in the movements within the ankle joint (Stecco, 2015). This is meaningful because it has been shown that pathologies of the Achilles tendon should not be seen in isolation but in context with the disposition of the plantar fascia. Therefore, the tightness of the plantar fascia has to be evaluated and, where necessary, treated in patients suffering from tendinopathies of the Achilles tendon (Chapter 14). High eccentric loading is considered the gold standard for treating pathologies of the Achilles tendon and, as pointed out previously, is a direct effect of fore-foot striking, which is promoted by barefoot running. Thus, biomechanical adaptations induced by barefoot locomotion are assumed to contribute positively to the treatment of tendinopathies of the Achilles tendon.

Patella tendon

The knee joint is another area of high axial loading, especially the Patella tendon, as the structure bearing the high loads generated by the quadriceps femoris plays a vital role in force distribution during locomotion. Interestingly, the Patella tendon is surrounded by a complex network of fascial tissue. In animal cadaver studies, it was shown that 15–20% of the longitudinal force at the knee joint is not transferred through the Patella tendon but through the surrounding fascial network (Yousefi et al., 2013). This finding draws attention within the context of footwear as footwear changes movement patterns not only at the ankle but also at the knee region, and thus intensely affects the fascial network surrounding the knee. In his study of the biomechanical effects of barefoot running on the knee joint, Sinclair (2014) concludes that the change

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in loading response might attenuate the risk of running-related knee injuries.

Epimysial fascia

Besides these distinct local fascial structures being affected, effects on the whole and complex fascial network, especially epimysial fascia, can be expected. Epimysial fascia surrounds and permeates muscles and is considered to contribute majorly to load transfer within and between muscles (Zügel et al., 2018). Similar to the function of the plantar fascia, epimysial fascia distributes and transfers load longitudinally. This aspect is of high importance in the foot, shank and thigh, as during locomotion high loads are transferred along the long axis of the leg. As illustrated previously, the fore-foot strike pattern causes high eccentric loads that have to be absorbed by the muscles of the foot and shank and, consequently, by the epimysial fascia, too. Recent findings suggest that micro-tearing of epimysial fascia and the presence of free nerve endings may contribute to the phenomenon of delayed onset of muscle soreness, often following the application of eccentric training stimuli (Chapter 16). Research of these mechanisms, however, are at an early stage and further progress should be expected in the near future.

Muscle septa

In addition to the aforementioned fasciae that contribute to the longitudinal load distribution, there is also stiff aponeurotic fascia tissue (septa), whose main function is the separation, connection and shaping of muscles and muscle groups from each other (Figure 32.6). Especially at the shank, these compartments need to be included in training considerations, as changes in locomotion are known to affect them. Namely, the interosseous membrane of the leg, the anterior intermuscular septum, the transverse

intermuscular septum and the posterior intermuscular septum, which divide the muscles of the shank into the anterior, lateral, deep posterior and superficial posterior compartments.

While these septa may not, or may only very slightly, be affected by the footwear type worn, they play a vital role, as sudden changes in footwear can result in painful consequences. The high eccentric loading of the shank muscles caused by fore-foot striking creates an effective training stimulus resulting in muscle hypertrophy, an increase in muscle thickness. Muscle hypertrophy is a mechanism happening at a fast rate, exceeding the rate the fascial tissue of the septa can adjust. Thus, the growing muscle within the septa induces compression forces that affect blood vessels and nerves traveling within the septa. This phenomenon is referred

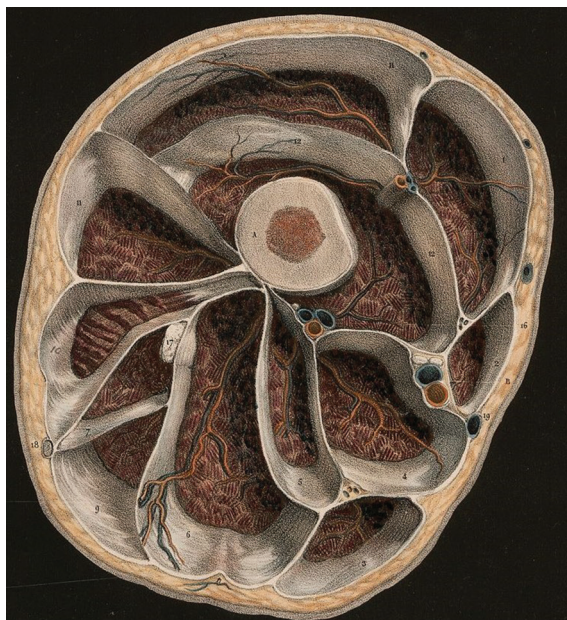


Figure 32.6

Muscle septa of the thigh shape the muscle compartments (Bourguery, 1832).

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to as chronic compartment syndrome and creates painful sensations due to nerve irritations and oxygen undersupply of the muscle tissue. To avoid the onset of a chronic compartment syndrome as a consequence of sudden changes towards barefoot or minimization footwear, a slow transition from regular footwear towards barefoot or minimization footwear is advised. Such a slow transition will allow the fascial tissue of the septa sufficient time to adjust to the growing muscle volume.

Movement coordination

Alongside the biomechanical aspects referred to, and linked to force absorption and force transfer, physiological sensation and subjective perception are worthwhile aspects to examine. There is a lack of studies specifically addressing the effects of footwear and their influence on physiological sensation and subjective perception within fascial tissue. There are, however, possible scenarios where footwear, or the absence of footwear, might alter subjective perception. In general, fascial tissue is highly innervated as various types of free and encapsulated nerve endings, including Ruffini and Pacinian corpuscles, are found inside. In particular, the epimysial fascia and the fascial network surrounding the Achilles and Patella tendons may contribute to the subjective perception of body posture and movement coordination. Running barefoot or in minimization footwear shifts the load absorption during foot strike from bony tissue towards the fascial tissue, as well as tendons and muscles of the foot and shank. Consequently, force absorption is directly connected to the tension of these tissues, while higher impact forces lead to higher tension. The highly innervated fascia network may therefore contribute to an increased awareness of changes in loading circumstances and joint angle position and can support movement

coordination control (Altman and Davis, 2012). Anecdotal reports of increased body feeling and control when running barefoot underpin the proposed effect.

Recommendations for individuals, coaches and therapists

While beneficial training effects on the fascia network and single fasciae have been illustrated, we need to be aware that no training magically changes body composition from weak to strong or from painful to pain-free. This also applies to the stimulation induced on the fascial network or single fasciae. Furthermore, while individuals, athletes and patients often have profound sensation of the disposition of bones, ligaments, muscles and tendons, less is known regarding their sensory awareness of their fascial network. While they clearly perceive better muscular strength relating it to the core contractile element of the muscle, this is different with regard to the tensile fascia components during muscular activity. Therefore, the success or failure of fascial training in healthy individuals and athletes is difficult to estimate due to inconclusive subjective perception and the lack of practical objective measures for respective validation. This differs when referring to patient groups, who are treated when subjectively experiencing inflammation and/or stiffening of the fascial network or single fasciae, frequently accompanied by perceivable discomfort or pain. The vanishing of such discomfort or pain then marks therapeutic success, also leading to an objectively measurable more efficient locomotion, alongside the regained practical ability to participate in standard physical exercise regimens.

Regarding training effectiveness, the dose of the stimuli applied to individuals, athletes and patients needs to be considered. While doses

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below a certain threshold will not result in bio-positive fascial tissue adaptation, applying doses above a certain threshold may result in an overload to the fascial tissue, which, similar to the more prominent passive (bones, cartilage and ligaments) and active (muscles and tendons) human skeletal structures, may be damaged as well. In this respect, the general training advice to increase training stimuli slowly and in response to the adapted tissue should prevail. When meeting these general recommendations, the likelihood to acquire a new and persisting injury are relatively small (below 3%) according to a study on 509 runners (Hryvniak et al., 2014). Furthermore, when teasing out further beneficial effects, 69% of the runners of that same study reported that their previous injuries vanished after switching to barefoot running. Research is also necessary to examine to what extent the fascial network adapts to the above training mechanisms, and to get an idea of threshold levels separating bio-positive from bio-negative fascial tissue adaptation.

Ultimately, it is not as simple as whether wearing shoes (or not) triggers beneficial structural and functional effects on the fascial network. It rather needs to be considered which type of shoe has which kind of effect. Based on the training or therapy objectives of individuals, athletes or patients, respective treatments should be executed, and certain shoes should be worn or taken off. In practical terms the use of instability footwear is recommended to stimulate the fascia network of the lower limb, especially the fascial network surrounding, crossing or inserting at the ankle region. By contrast, instability footwear, which sometimes appears bulky due to added material to the midsole and outsole, does not overly stimulate the intrinsic fascial tissue of the foot. Thus, for the latter, minimization footwear usage or barefoot locomotion is

recommended, which can be complimented by exercising on soft surfaces.

Finally, while it is easy to observe what type of shoe is worn or which type of surface locomotion and exercise are carried out on, this does not allow us to judge whether the physical activity executed is adding structural or functional stimulation to the fascial network. Such assessment needs to be based on the actual execution of walking, running or other training exercises. Using minimization shoes or moving barefoot will only additionally stimulate the fascial network when the foot-strike behavior is considerably changed and mid-foot or fore-foot strikes are applied, as otherwise the force attack location remains at the bony calcaneal region. In this sense, minimization footwear and barefoot locomotion can act as a trigger to execute strike types beneficial for fascial tissue; however, these are not the purpose and effect by themselves. It has been shown that when barefoot, people often still apply rear-foot strikes, just in a rather moderate style (Sterzing et al., 2010).

Conclusion and perspective

This chapter has introduced three mechanisms of locomotion and exercise adaptation assumed to be beneficial for bio-positive stimulation of the fascial network and single fasciae of the lower extremity. These mechanisms are fostered by wearing different types of shoes or by not wearing shoes at all. Namely, they are (1) a distal shift of foot-strike location during walking and running triggered by minimization footwear or barefoot activities, (2) an increase in intrinsic muscular and fascial tissue activity present during barefoot locomotion, (3) an increase in ankle joint movement triggered by instability footwear. The topic of training stimulation of the fascial network by footwear is innovative,

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therefore this chapter has a derivative nature. The combination of future practical experience and scientific research is needed to verify the proposed mechanisms and, furthermore, to subjectively and objectively quantify respective structural and functional effects on the fascial network and single fasciae when incorporated in training regimens. For evidence-based guidance of individuals, athletes and patients, as well as of their respective coaches and therapists, to induce positive footwear-related effects on fascial tissue, the following information is needed: type of footwear, type of movement, time period of stimulation, and also certain characteristics of the individual. To date, sound scientific evidence is lacking to provide most of this information.

The effects of barefoot running, and especially the effects of mid- and fore-foot running patterns, on muscles and tendons, are already known (Altman and Davis, 2012). These effects will likely be transferable to the fascial network surrounding these structures. Thus, the

eccentric training stimulus created by mid- to fore-foot strike patterns that are usually induced by barefoot locomotion exposes the fascia tissues of the leg, especially the foot and posterior shank, to high loading stress. This seems to provide the necessary stimulus for the fascial tissue to increase the number of collagen fibers and to adjust their alignment towards the applied force vector. Both adaptations increase the stiffness of the fascial structures and may contribute to foot arch stabilization and treatment of various pathologies of the plantar fascia, Achilles tendinopathy, and anterior knee pain. While we see these potential positive effects promoting at least a sporadic, if not general, inclusion of barefoot or minimization footwear locomotion into the training regime of athletes and physically active people, it has to be pointed out that research in this area is in its infancy and that future research is mandatory to provide therapists and coaches with the necessary information for standardized applications.

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